

Griffin Halloran

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EDUCATION

Indiana University - Bloomington
Master of Science in Computer Science
Bachelor of Science in Computer Science

Bloomington, IN
December 2020
December 2019

EXPERIENCE

Autonomous Software Engineer

Feb. 2021 – Present

L3Harris

- Developed C++ software for autonomous surface vessels, supporting self-driving and unmanned ocean vessel operations
- Designed and implement a multi-heuristic variant of the A* pathfinding algorithm, enabling configurable AI behaviors through interchangeable heuristic functions
- Implemented software for sensor fusion and perception systems, utilizing TCP and UDP networking to transmit and integrate sensor data across autonomous vehicle applications

Robotic Process Automation and Machine Learning Intern

June 2020 – August 2020

Caterpillar Inc.

- Applied image-processing and machine learning techniques to segment and categorize handwritten check data
- Developed machine learning solutions to automatically detect, classify, and route invoices across the company
- Automated repetitive business processes by creating Robotic Process Automations (RPAs) for company-wide workflows

Department of Navy Pathways Internship

Aug. 2017 – Aug. 2019

Naval Sea Warfare Center, Crane

- Developed and tested ROS-based robotic systems for the Platform Engineering Branch, working with robotic software, sensors, and autonomous system capabilities
- Helped establish a machine learning research laboratory to support on-base research and experimentation in artificial intelligence
- Created an object detection program to identify and classify objects captured from cameras

TECHNICAL SKILLS

Languages: C++, C#, Python, C, Java

Technologies: ROS 2, Docker, DDS, Linux, TCP/UDP, CMake, Qt

Tools: Mercurial, Git, Jira, Confluence, Visual Studio, Vim, LaTeX

PROJECTS AND RESEARCH

Indiana University, Computer Vision Lab - Research Assistant

Jan. 2019 – Dec. 2020

- Researched architectural techniques used to design accelerators for neural networks to reduce network computational costs
- Investigate visualization methods for machine learning practices to gain a better understanding of debugging practices